MILITARY SPECIFICATION

CONTROL, RADIO SET C-6684(*)/ASQ

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide control of the following functions when installed in the Integrated Electronic Centrals AN/ASQ-88A or AN/ASQ-19A.

(1) UHF Communication Equipment

(2) UHF Aux Receiver Equipment

(3) UHF ADF Equipment

(4) Comm Channel Indicator ID-1311/ASQ.

1.2 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.4.

1.3 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents, unless specified in the contract or order (see paragraph 6.8).

2. APPLICABLE DOCUMENTS

2.1 General - The documents listed below of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

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2.1.1 Applicable Documents -

Military

Mil-I-18309	Integrated Electronic Central AN/ASQ-19A
MIL-I-81036	Integrated Electronic Central AN/ASQ-88A
MIL-C-6781	Control Panel, Aircraft Equipment, Rack or Console Mounted
MIL-P-007788C	Panels, Information, Integrally Illuminated

2.1.2 Availability of Documents

(1) When requested specifications, standards, drawings, and publications refer to both title and number, copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.

3. REQUIREMENTS

3.1 Preproduction - Preproduction test requirements shall be in accordance with AN/ASQ-19A specification MIL-I-18309.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the primed consideration. In so doing, parts and materials shall be in accordance with AN/ASQ-19A specification MIL-I-18309.

3.3 Design and Construction - This equipment shall conform with all applicable requirements of AN/ASQ-19A specification MIL-I-18309 for design, construction, and workmanship, except as otherwise specified herein.

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3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 6.0 pounds.

3.3.2 Reliability

3.3.2.1 Operating Life - The equipment shall have total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.2 Reliability in Mean Time Between Failures (MTBF) - The equipment shall have a specified mean (operating) time between failures of 1000 hours when tested and accepted as outlined in AN/ASQ-19A specification MIL-I-18309.

3.3.3 External Connectors - The two external connectors shall be an MS 3112E-20-41P and MS 3112E-18-32PW mounted at the rear of the equipment.

3.3.4 Lighting Panel - The lighting panel shall conform to the requirements of MIL-P-007788C, class 1-R type V.

3.3.5 Standard Conditions - The conditions used to establish normal performance characteristics under standard conditions and for making laboratory bench tests shall be in accordance with AN/ASQ-19A Specification MIL-I-18309 except for the following:

Input Power Voltage 33 + 2, -3 V DC

3.3.6 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in specification MIL-I-18309 for Class I equipment.

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3.3.6.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve I of Specification MIL-E-5400.

3.3.7 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-I-18309. The power requirements shall not exceed the specified amounts.

(1)	DC Power 28 volts (panel lights)	.320 amp
(2)	DC Power 28 volts	.110 amp
(3)	DC Power 28 volts (to ID-1311 Indicator)	.200 amp

NOTE: Item (3) will be .600 amp when ID-1311 is channeling.

(4)	DC Power 33 volts (from RT-793/ASQ)	.500 amp.

3.4 Performance

3.4.1 General - Unless otherwise specified, values set forth to establish requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.2 UHF Communication Portion of the C-6684/ASQ

3.4.2.1 Frequency and Channel Selection - Manual and preset frequency selection is provided for in the range 225.00 to 399.95 mc. A preset channel solid state memory device shall be provided for channel storage that can be reset from the front of the control. Provision is made for one or more channel indicators to be used in conjunction with the equipment.

3.4.2.2 Volume Control - Operation of the volume control shall provide for adjusting the amplitude of the audio signals delivered to the output circuit from the UHF Communication equipment.

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3.4.2.3 Function Control - Operation of the function selector shall permit selection of the following modes of operation on the UHF Communication equipment:

(1) Transmit and receive.

(2) Transmit and receive plus Guard.

(3) ADF plus Guard.

(4) ADF.

Green low level illumination of the selected pushbutton shall be provided for night time viewing. This control shall operate in conjunction with the UHF auxiliary Receiver and the UHF ADF selector controls. (See 3.5.1.3.4).

3.4.3 UHF Auxiliary Receiver Portion of the C-6684/ASQ

3.4.3.1 Channel Selection - Channel selection is provided for any one of twenty available channels.

3.4.3.2 Volume Control - Operation of the volume control shall provide for adjusting the amplitude of the audio signals delivered to the output circuit from the UHF Auxiliary Receiver equipment.

3.4.3.3 Function Control - Operation of the function selector control shall permit selection of the following modes of operation on the UHF Auxiliary Receiver Equipment:

(1) ADF

(2) Preset or Command Channels

(3) Guard.

This control shall operate in conjunction with the UHF Communication and UHF ADF selector controls. (See 3.5.1.3.4).

3.4.4 UHF ADF Portion of the C-6684/ASQ

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3.4.4.1 Function Control - Operation of the function selector control shall permit selection of automatic direction finding operation. This control shall operate in conjunction with the UHF Communication and UHF Auxiliary Receiver selector controls. (See 3.5.1.3.4).

3.4.5 Dual Operation - Operation of the COMM CMD and NAV CMD controls shall permit either operator to "take control" separately over the navigation and communication functions.

3.5 Detail Requirements

3.5.1 General - The equipment shall be of the console type in accordance with Specification MIL-C-6781, and shall fulfill the following requirements.

3.5.1.1 Function - The equipment shall provide control of the UHF Communication, UHF Auxiliary Receiver, and UHF ADF equipment portions of the Integrated Electronic Central AN/ASQ-19A or AN/ASQ-88A. It shall be possible to select manually any one of the 3500 frequencies available on the UHF Communication portion and to preset up to 18 channels on any one of these available frequencies. In addition, it shall be possible to select a fixed tuned guard channel. Unless otherwise specified, the equipment shall be supplied with the guard frequency set for 243.00 mc. The equipment shall also provide information for remote indication of the UHF Communications preset channels (1 through 18), guard channel, and manual operation. Provisions shall be available for dual operation control of the UHF Communication, UHF Auxiliary Receiver, and UHF Navigation portions of the AN/ASQ-19A. It shall be possible also to select any one of twenty UHF Auxiliary Receiver channels and to select operation of the UHF ADF equipment when desired.

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3.5.1.2 Form Factor - The equipment dimensions shall conform to the applicable requirements of Specification MIL-C-6781 for a Type I control panel and shall not exceed 6 3/8 inches in length and 5 inches in depth. Projecting knobs shall not exceed 1 1/4 inches from the mounting surface.

3.5.1.3 Controls - The following controls shall be provided and suitably marked on the face of the equipment.

3.5.1.3.1 UHF COMM -

(1) Tuning Mode Selector - A three position toggle shall be provided for selecting channel, guard, or manual modes of operation. In the channel position, frequency control originates at the preset memory and the remote indicators are controlled to display the channel number (1 through 18). In the guard position, frequency control originates at the guard frequency terminal board (see 3.5.1.3.6) and the remote indicators are positioned to display "G". In the manual position, the frequency control originates at the frequency selectors and the remote indicators display "M". In addition, the toggle shall operate a shutter on the COMM channel digital display. The shutter will cover the display in the guard and manual mode and uncover the display in the channel mode.

(2) Channel Selector - A rotary knob with a separate digital readout shall be provided for the selection of eighteen (18) preset frequencies.

(3) Frequency Selectors - Four linear actuators with a separate digital readout shall be provided for manually selecting any one of 3500 available frequencies from 225.00 mc to 399.95 mc in .05 mc increments. The actuators shall be moved down for increasing numbers.

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(4) Channel Preset Control - A push button shall be provided for changing the frequencies set into the preset memory. When the button is actuated with the tuning mode selector in CHAN position, the frequency set on the frequency selectors will be inserted into the channel set on the channel selector. Actuation with the mode selector set to GUARD or MANUAL will cause no change. The push button shall be recessed below the surface of a surrounding guard to minimize accidental operation.

(5) Volume Control - A linear actuator shall be provided for adjusting the amplitude of the audio signals delivered to the output circuit from the UHF Communications equipment. Operation of the volume control shall cause a minimum of change in the impedance which the radio set presents to the external equipment attached to the audio output terminals. The characteristic impedance of the attenuator shall be nominally 600 ohms at any setting of the volume control. The control shall have a detented OFF position which disables the COMM and AUX functions. The actuator shall have numerals 0 through 9 on the face to indicate relative volume settings. The numeral 0 shall be centered for the OFF position and increasing numerals shall indicate increasing volume. The numerals shall not be lighted.

(6) Function Selector - See 3.5.1.3.4.

3.5.1.3.2 UHF AUX REC -

(1) Channel Selector - A rotary knob with a separate digital readout shall be provided for selection of twenty UHF Auxiliary Receiver Channels.

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(2) Volume Control - A linear actuator shall be provided for adjusting the amplitude of the audio signals delivered to the output circuit from the UHF Auxiliary Receiver equipment. Operation of the volume control shall cause a minimum of change in the impedance which the radio set presents to the external equipment attached to the audio output terminals. The characteristic impedance of the attenuator shall be nominally 600 ohms at any setting of the volume control. The actuator shall have numerals 0 through 9 on the face to indicate relative volume settings. The numeral 0 shall be centered for the OFF position and increasing numbers shall indicate increasing volume. The numerals shall not be lighted.

3.5.1.3.3 UHF ADF -

(1) Function control shall be provided as described in paragraph 3.5.1.3.4.

3.5.1.3.4 Function Selector - Four interlocked push buttons shall be provided for the selection of the following modes of operation of the UHF Communication, UHF Auxiliary Receiver, and UHF ADF equipment:

POSITION	UHF COMM MODE	AUX REC MODE
1 (left button down)	T/R	ADF
2	T/R + G	ADF
3	ADF + G	CMD
4	ADF	G

3.5.1.3.5 Take Control Selectors

(1) A push type "Take Control" switch shall be provided for NAV CMD operation when dual controls are installed. This switch controls an external relay (or relays as necessary) to switch control functions as desired. An associated light is controlled

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by the external relay (or relays) and will be illuminated on the operator's panel that has control. This control also functions as a "give" control if desired. Color of lights shall be identification green conforming to FED-STD-3.

(2) A push type "Take Control" switch shall be provided for COMM CMD operation identical in operational characteristics to (1) above. In addition, the control shall be designed such that the COMM frequency control output shall be disabled by switching one wire at the external relay. Disabling the frequency control output shall not cause the channel set capability (see 3.5.1.3.1) to be disabled.

3.5.1.3.6 Controls, Internal

(1) A preset channel solid state memory device shall be provided. The stored information shall not be affected by power interruptions, environmental conditions, or storage for long periods of time. Channel information that has been stored previously shall be erased upon selection of a new preset channel frequency.

(2) A jumper terminal board shall be provided internally for setting the guard frequency. Unless otherwise specified, the guard frequency shall be set for 243.00 mc.

3.5.1.3.7 Audio Circuits - Audio load resistors of 620 ohms, 1 watt, shall be connected from each of the audio output terminals of the equipment to a separate pin on the connector, where they may be individually grounded at the option of the user.

4. QUALITY ASSURANCE PROVISIONS - The equipment shall conform to the Quality Assurance provisions of MIL-I-18309 except as otherwise specified herein.

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4.1 Manufacturing Run In Test - At the completion of the specified vibration period, each control of the equipment shall be operated for ten complete operations without failure.

4.2 Scope of Sampling Tests - Scope of the tests shall be as specified in specification MIL-I-18309, except that Manufacturing Run In Test shall consist of 200 operations of each control with no restrictions on the number of failures.

4.3 Life Test - Life tests of the equipment shall consist of 10,000 operations each of the COMM manual tuning controls and 25,000 operations of the COMM and AUX channel selectors.

5. PREPARATION FOR DELIVERY

5.1 General - Preparation for delivery shall be in accordance with MIL-I-18309.

6. NOTES

6.1 Intended Use - This equipment is intended to be used to provide control of the UHF Communication, UHF Auxiliary Receiver and UHF ADF equipments of Integrated Electronic Central AN/ASQ-88A or AN/ASQ-19A.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V DC and 33 + 2, -3 V DC as required by the equipment. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

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(2) Central Specification - Integrated Electronic Central AN/ASQ-19A Specification MIL-I-18309 shall have precedence over this specification.

(3) This Specification - This specification shall have precedence overall applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(4) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Associated Equipment - The equipment shall operate with the following associated equipment:

ITEM	TYPE DESIGNATION
Receiver-Transmitters Radio	RT-793/ASQ
Amplifier-Power Supply-Receiver	AM-2349/ASQ-19
Antenna	AS-909/ARA-48
Intercommunications Station	LS-460/AIC
Intercommunications Station	LS-459/AIC
Indicator, Frequency Channel	ID-1311/ASQ

6.5 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the

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use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.6 As a general rule non-repairable, subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.7 The parentheses (*), when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with AN/ASQ-19A Specification MIL-I-18309. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.8 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

(3) Data requirements (see 1.3).

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-C-81441(AS) CONTROL, RADIO SET C-6684(*)/ASQ

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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